PYTHON

VARIABLES

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https://thehonestcoder-edu.github.io/

ANALOGY

Indian Family Example

We buy commodities like Ghee and Surf. Once those items get finished, we reuse them wonderfully. You'll never realize that the jar titled 'Pure Cow Ghee' could ever contain spices! Or pickles! Or even sweets!

Even though the basic container (Ghee jar) remains the same, the contents keep changing(when you switch it of course!)

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What are variables?

Variables, according to our analogy, are the Ghee jars, or in other words, the container which stores items. And the contents are the values associated with the variable.

VARIABLES

VARY ABLE

Going by the word composition, variable literally means ‘able to vary’. Thus, it means that its values can be changed.

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Syntax-

variable = value

For example, if i want to say that the user's age is 32, i can do it as follows -

age = 32

Reassignment

Now, let us say I want to update the age as 33 years. I can do it as follows:

```
age = 32
age = 33
```

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Walking through the Code

Line 1: The computer creates a box ‘age’ and sets it to 32.

Line 2: It looks into its memory - “Hmm...Is there any box in my memory named ‘age’?”

Upon searching it finds a box containing the number ‘15’. But since we say that age is now equal to 16, the computer says -

“Never Mind that ‘15’. The user said ‘16’, so it is ‘16’.” And it now stores the number ‘16’ in that storage box.

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Variable Naming Conventions

These are practices followed worldwide for overall code readability. Some of these are as follows:

- Use all lowercase letters with underscores between words.
- Always start with a letter instead of a number.
- Keep variable names short yet meaningful.
- Avoid reserved keywords.

But, what *are* keywords?

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#Definition

Reserved Keywords

These are "System Words" that Python has already claimed for itself. They are like the "Reserved" signs on tables at a restaurant. These words already have 'Jobs' in Python.

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#Definition

Reserved Keywords

If you try to say print = 5, you've just fired the person in charge of displaying text! Python would let you do that, but you have broken some serious functionality out there! And in some strict cases, python would simply crash out!

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#Pro Tip

How to Identify?

Don't stress about memorizing a long list of words!

Watch the Colors: Your code editor will highlight these words (usually in Purple or Orange). If the color changes, the word is "Taken."

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Getting the contents out

To display anything, we used the print() function. The same goes here!

```
age = 16
print(age)
```

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#Pro Tip

Common Mistake - Avoid!

If the mentioned variable does not exist, Python throws an error.

Common Mistake - Avoid!

Python is case-sensitive. Thus, ‘age’ is different from ‘Age’. Make sure!

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Quotes V.S No-Quotes

Quotes

- Python treats "age" as text.
- It prints the text exactly.
- Output: age

No Quotes

- Python treats it as a variable.
- It looks for the variable and prints its value.
- Output: 16

Key Concept

- "age" → text
- age → variable

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#Pro Tip

Remember

There aren't any real boxes hiding inside your laptop! It just refers to a portion of memory.

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